Computer Networks

Lecture-2

Recap of Lecture 1

- Data Communication
- Data Communication System
- Factors effecting Data Communication system
- Components of data communication system
- Data Representation

Overview of Lecture 2

- Key Data Communication Terminology
- Data Flow types
- Networks and Why we need Them?
- Distributed Processing
- Network Criteria
- Physical structure of Network
  - Type of connection

Key Data Communication Terminology

- Session
  - Communication dialog between network users or applications.
- Network
  - Interconnected group of computers and communication devices.
- Node
  - Any device that is part of network.
  - Any device that is capable of sending or receiving data (e.g. computer, printer etc.)
- Link
  - connects adjacent nodes.
  - e.g. Wires, Cables, any thing that physically connects two nodes, or it can be wireless.

Key Data Communication Terminology

- Path
  - end-to-end route within a network.
- Circuit
  - the conduit or physical path over which data travels.
- Packetizing
  - Dividing messages into fixed-length packets prior to transmission over a network's communication media.
- Routing
  - Determining a message's path from sending to receiving nodes.

Data Flow / Transmission Mode

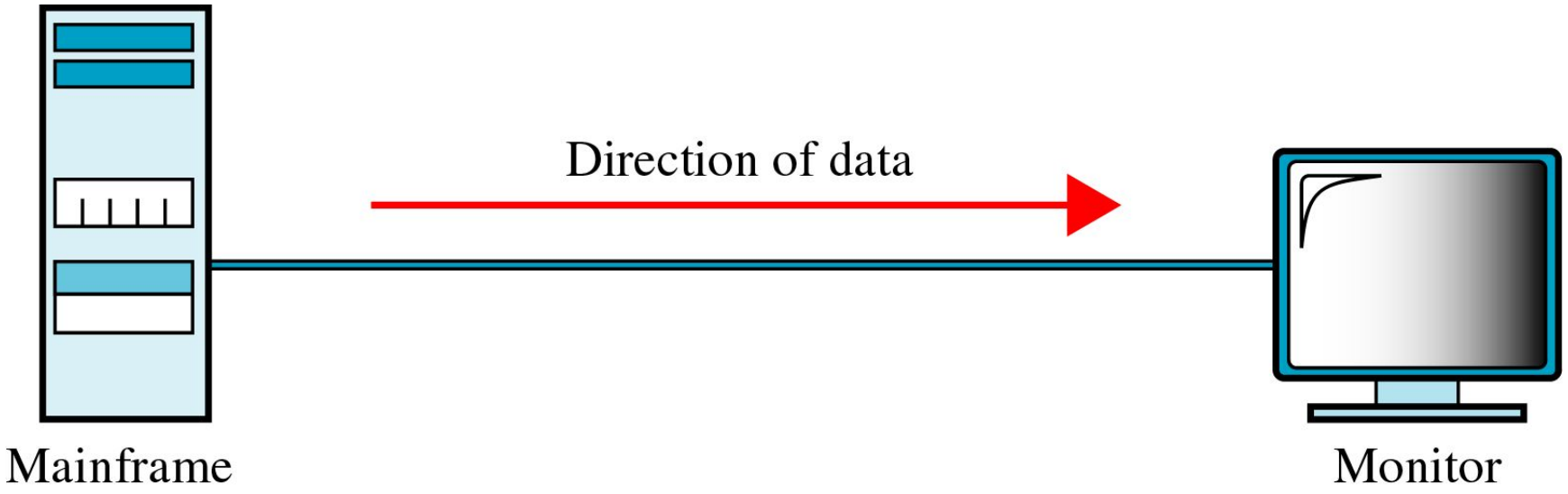
- Communication between two devices can be;

1. Simplex
2. Duplex
   - Half-duplex
   - Full-duplex

Simplex

- Communication take place only in one direction.
  - Only sender just sends data & receiver just receive data.
  - The simplex mode can use the entire capacity of the channel to send data in one direction.
  - Called as one-way transmission
    - i.e. there is no back direction of transmission from receiver to sender
  - e.g.
  - Radio, TV transmission.
  - Input provided to the computer by keyboard, mouse, or scanner etc.
  - The flow of data from the computer to monitor, and printer etc.

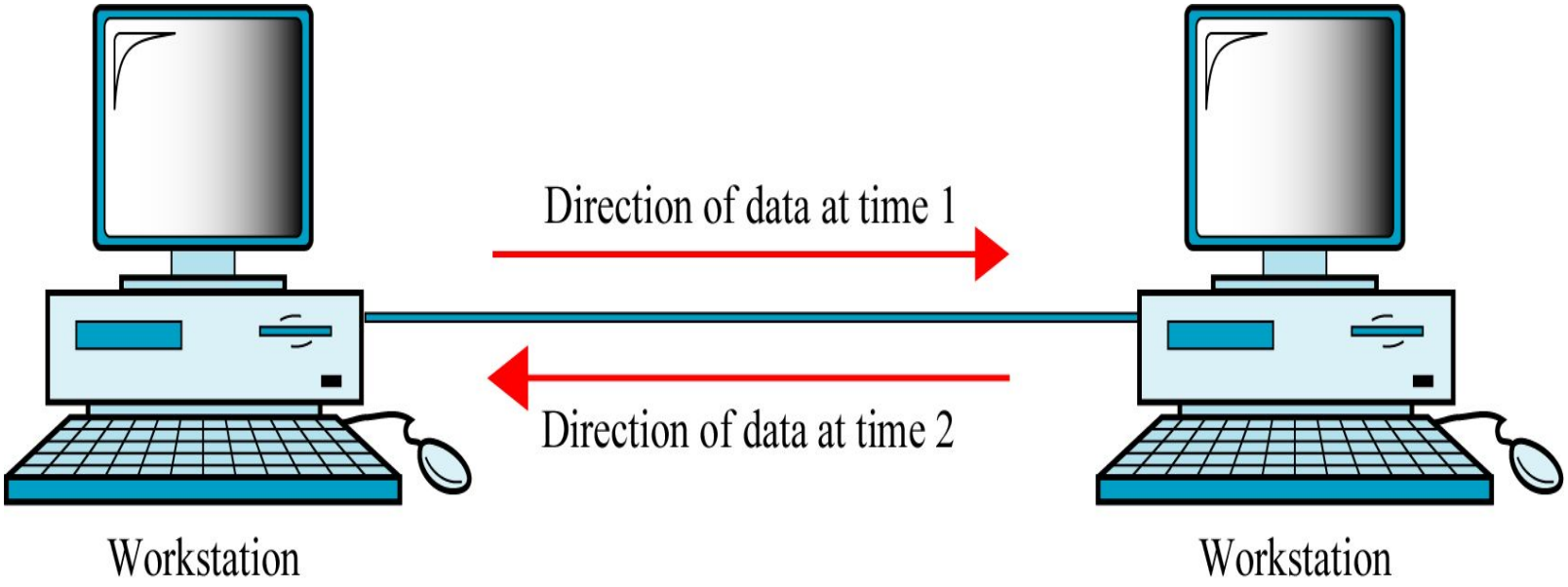
Simplex



Half-Duplex

- Each station can both transmit and receive, but not at the same time.
- When one device is sending, the other can only receive, and vice versa
- the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time.
- e.g.
  - wireless system
  - walky-talky
  - In computer field a multi-user environment is a best example of half duplex mode, in which a terminal transmits data to main computer and then the computer responds with acknowledgment.

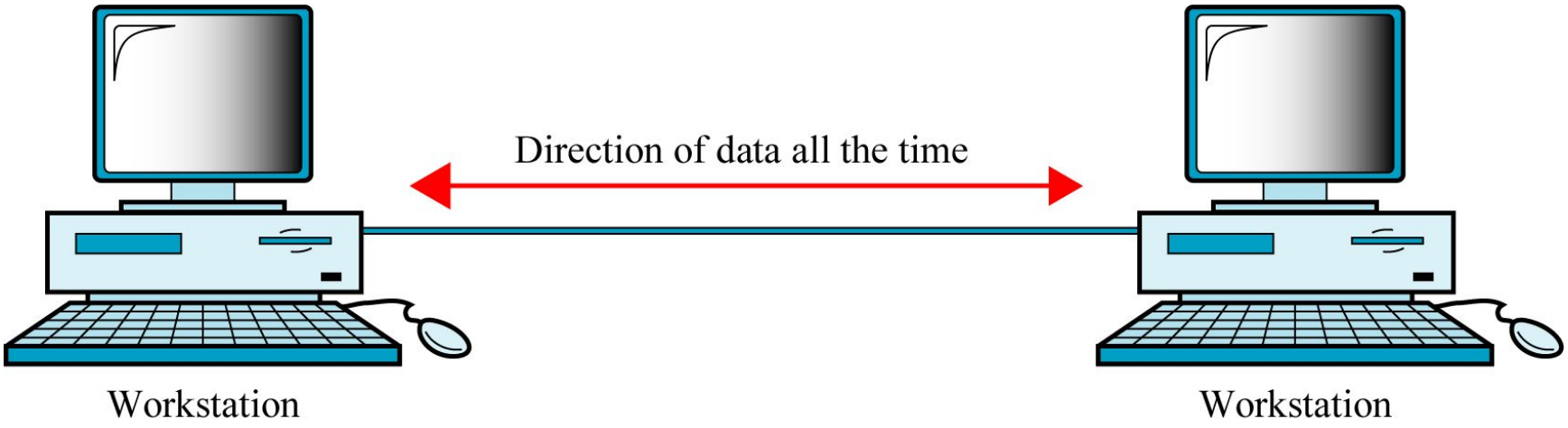
Half-Duplex


Direction of data at time 1
Direction of data at time 2
Workstation
Workstation

Full Duplex

- Two-way transmission.
- Both sender and receiver transmit data simultaneously.
- Signals going in either direction share the capacity of the link.
- This sharing can occur in two ways:
  - Either the link must contain two physically separate transmission paths
  - Or the capacity of the channel is divided between signals traveling in both directions.
  - The best example is telephone communication. Similarly modem card, NIC, sound card etc are the devices used in computer technology for full duplex transmission.

Full Duplex



Networks

Networks

- When two or more Devices (Nodes) are connected through a medium (e.g. cable) to share their resources and information, it is called a network".

- A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

- In other words, a group of computers and other devices connected together is called computer network, and this process is called computer networking.

Networks- Why we need Them?

- Point to point communication not usually practical
  - Devices are too far apart.
  - Large set of devices would need impractical number of connections.

- Solution is to connect all devices to a central system known as a NETWORK

Benefits Of Networking

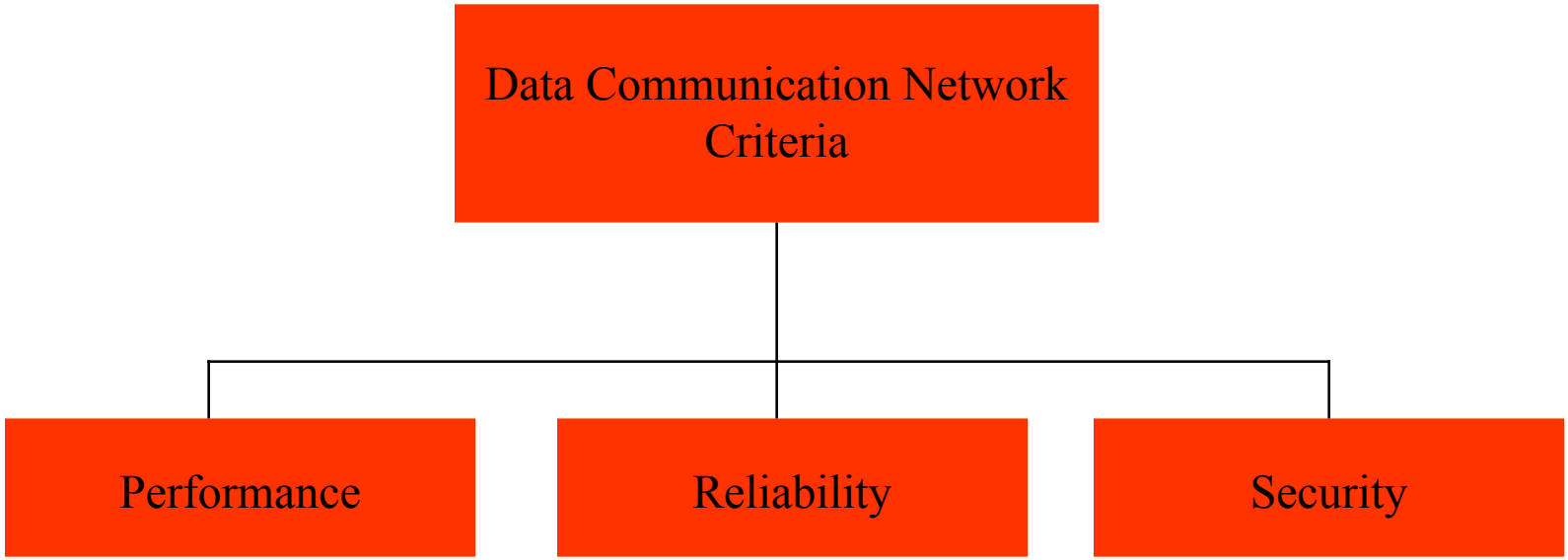
- Sharing of information.
  - e.g. sharing information in business environment
- Sharing the hardware resources.
  - E.g. Printer, modem, scanner, hard disk, CDRom, floppy disc, plotter etc.
- Sharing Software Resources.
  - MS Office and Lots of application software.
- Preserving the information.
  - Centralized backup.
- Protecting the information.
  - By applying authentication schemes.
- E-mail
  - people to communicate.

Distributed Processing

- “Instead of a single large machine being responsible for all aspects of a process, each separate computer handles a subset of the task”

- Most networks use distributed processing, in which a task is divided among multiple computers.

Network Criteria



Network Criteria

1. Performance

- Can be measured in many ways including Transit and Response time.
  - Transit time is the amount of time required for a message to travel from one device to another.
  - Response time is the elapsed time between an inquiry and a response.
- Performance depends upon these factors
  1. Number of Users
  2. Type of Transmission Medium
  3. Capabilities of Hardware
  4. Efficiency of Software

Network Criteria

2. Reliability

- Depends upon a no. of factors
  - Frequency of failure
  - Recovery Time of a network after Failure
  - Network's robustness in a catastrophe.
    - E.g. Fire earthquake or theft

Network Criteria

3. Security

- Issues include;
  - Protecting data from unauthorized access,
  - Protecting data from damage,
  - and implementing policies and procedures for recovery from breaches and data losses.

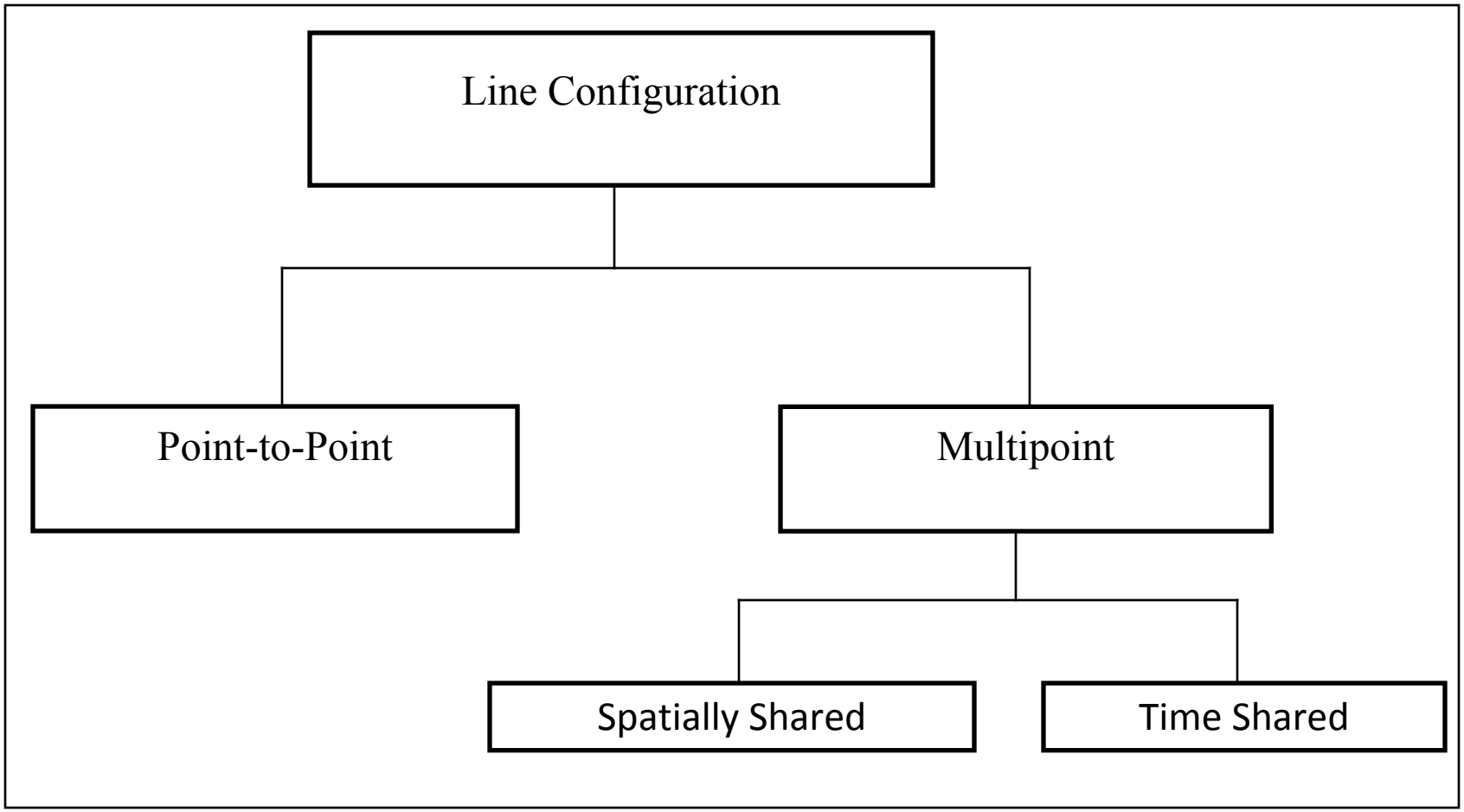
Physical Structures of Network

1. Type of Connection
2. Topology

Physical Structures

- **Type of Connection**
  - Also refer as Line Configuration.
  - “Line Configuration refers to the way two or more devices attach to a link”
  - In a network two or more devices are connected through links.
  - A link is a communications pathway that transfers data from one device to another.
  - For communication to occur, two devices must be connected in some way to the same link at the same time.
  - There are two possible types of connections:
    - point-to-point
    - and multipoint.

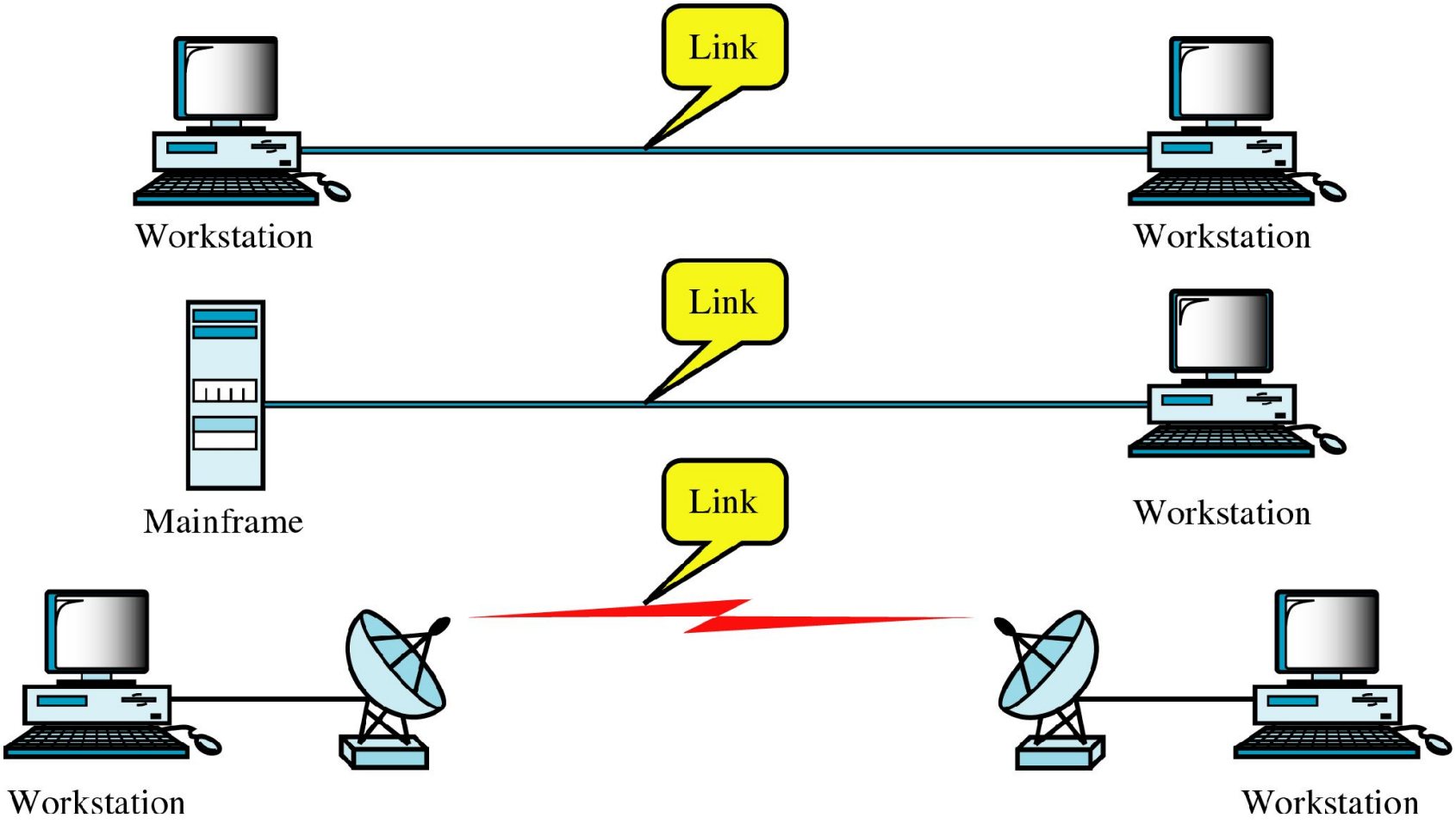
LINE CONFIGURATION



Point-to-Point

- A point-to-point connection provides a dedicated link between two devices.
- The entire capacity/bandwidth of the link is reserved for transmission between those two devices.
- Most point-to-point connections use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible.
- Data and control information pass directly between entities with no intervening agent.
  - e.g. TV Remote Control, mobile phone when talking, Base station (Antenna)

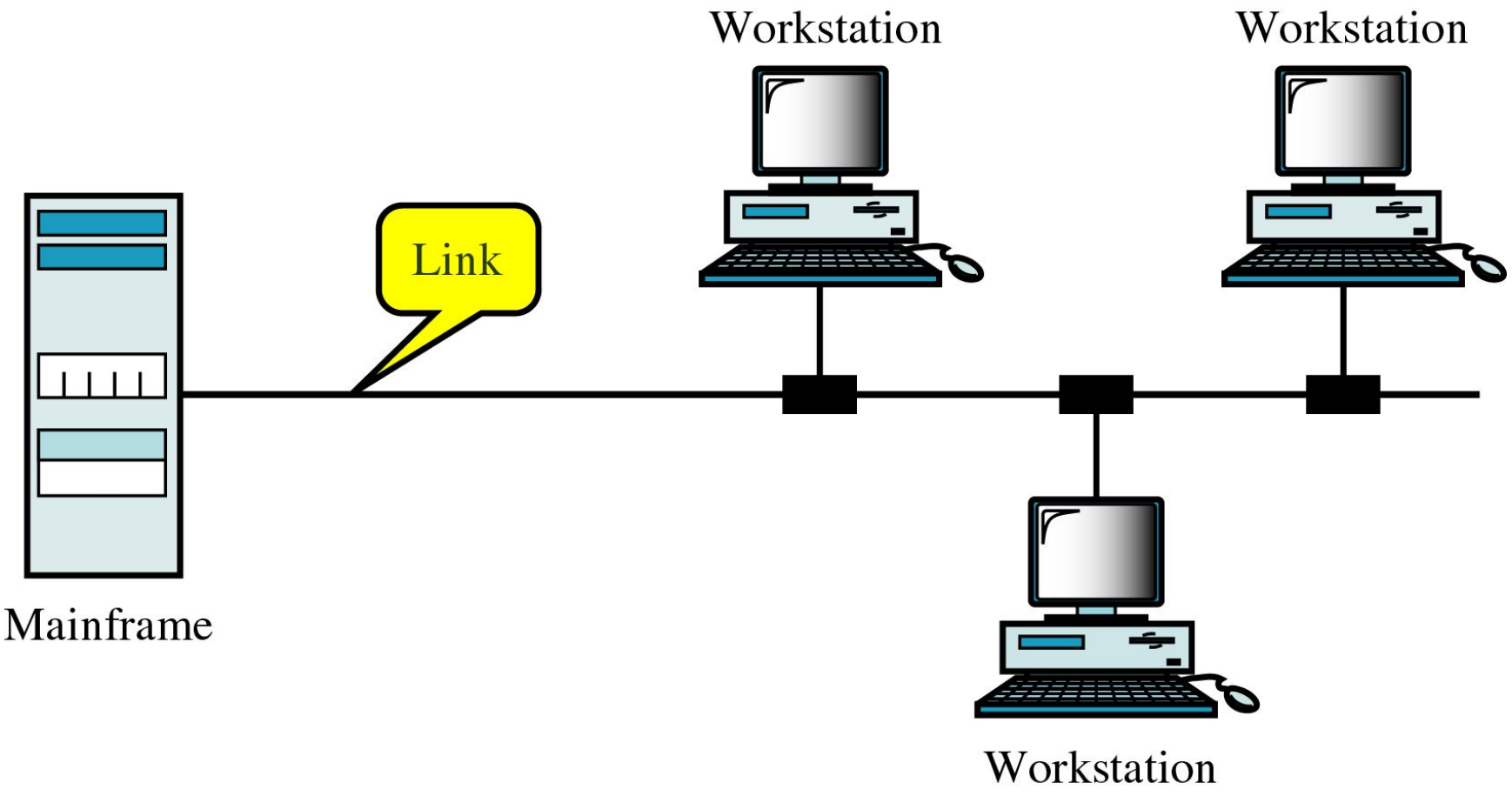
Point-to-Point Line Configuration



Multipoint

- A multipoint (also called multi-drop) connection is one in which more than two specific devices share a single link.
- Capacity (bandwidth) of the link is shared.
  - Spatial sharing
    - Multiple devices share the link simultaneously.
  - Temporal (Time) sharing
    - User uses the link only at their time slot.

Multipoint Line Configuration



Summary

- Key Data Communication Terminology
- Data Flow types / Transmission Modes
- Networks and Why we need Them?
- Distributed Processing
- Network Criteria
- Physical structure of Network
  - Type of connection

Suggested Reading

- Section
  - 1.1,
  - 1.2,
    - "Data Communications and Networking" 4th Edition by Behrouz A. Forouzan

- Sections
  - 1.3,
    - "Data and Computer Communication" 6th Edition by William Stallings